Qingyang Xiong

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GitHub
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Education

- **Nanjing University of Information Science & Technology** *2018-2022*
  ***B.S.** in Atmospheric Sciences* ***Advisor:** Juan Li* GPA: 3.0/4.0
- **Second Institute of Oceanography, MNR** *2022-2025*
  ***M.S.** in Physical Oceanography* ***Advisor:** Qiaoyan Wu* GPA: 4.0/4.0
- **City University of Hong Kong** *2025-Present*
  Ph.D. candidate ***Advisor:** Ping Lu*

Research Projects

- **Evaluation of CMIP6 Models' Performance on Simulating the Asian-Australian Monsoon.** *2021-2022*
  Undergraduate thesis
  - **Keyword: Python, SEOF, CMIP6**
  - Evaluated the performance of 13 climate models and their Multi-Model Ensemble (MME) in simulating the climatology of the Asian-Australian Monsoon.
- **The Relationship between Maximum Intensification Rate and Lifetime Maximum Intensity of TC.** *2023-2025*
  Main research topic of postgraduate
  - **Keyword: Carnot Heat Engine Theory, Differential Equation, WRF Ideal Experiment, Python**
  - Based on the Carnot engine theory and the intensification rate (IR) equation of tropical cyclone (TC). Primary focus on how the maximum intensification rate (MIR) of TC influences its lifetime maximum intensity (LMI). The discussion examines the timing and magnitude relationships. Validation of the conclusions is conducted best-track datasets and full-physics numerical simulation.
- **Modulation of Typhoons with Concentric Eyewalls in the Western North Pacific by ENSO.** *2025-2026*
  Tropical Cyclone Climatology
  - **Keyword: Tropical Cyclones, Concentric Eyewalls, ENSO, Climatology**
  - Using 30 years of concentric-eyewall observations (1991–2020), this study examines how the El Niño–Southern Oscillation (ENSO) modulates typhoons with concentric eyewalls in the western North Pacific and addresses the limited understanding of climate-variability controls on concentric-eyewall activity.

Skills & Interests

Programming: Python, LaTeX, Matlab, Fortran
Skills: Dynamic Meteorology, tcpyPI, WRF, Overleaf
Field of Interest: Atmospheric Dynamics, Tropical Cyclone Dynamics, Data science
Language Proficiency: TOEFL iBT: 81
Hobbies: Basketball, Physics, Reading

Achievements

- **Full marks (4.0/4.0) in Dynamic Meteorology at undergraduate (92/100) and graduate level (100/100)**
- **First class scholarship for graduate students, SIO** *2023*
- **Second class scholarship for graduate students, SIO** *2024*
- **Oral presentation at Xiamen Symposium on Marine Environmental Sciences, Xiamen** *2025*
- **Excellent oral presentation at The 2nd Future Atmospheric Science Forum, Suzhou** *2026*

Publications

Modulation of Typhoons with Concentric Eyewalls in the Western North Pacific by ENSO

Authors: Qingyang Xiong, Ping Lu, and Pak Wai Chan.

Submitted to *Journal of Climate*, 2026.

Abstract. Concentric eyewalls (CEs) occur in over 70% of major tropical cyclones (TCs) and can be significantly influenced by large-scale environments. Despite extensive research linking climate variability to general TC characteristics, studies on the modulation of CEs by climate variability are quite limited. This study addresses this gap using 30-year CE observations (1991–2020) and examines how the El Niño–Southern Oscillation (ENSO) modulates typhoons with CEs in the western North Pacific (WNP). Results show that ENSO modulates CE TCs primarily by regulating the occurrence of major TCs, with more major TCs and CE TCs during the Warm phase. Among major TCs, the fraction of CE TCs is also higher during the Warm phase, suggesting an ENSO contribution to CE formation beyond its influence on TC intensity. Compared with major non-CE TCs, major CE TCs tend to occur under higher sea surface temperature, larger ocean heat content (OHC), more favorable convective conditions, greater atmospheric moisture, weaker vertical wind shear (VWS), and larger offshore distance. CE TCs in Warm phase are further characterized by weaker VWS, larger OHC, and greater offshore distance than their counterparts in other phases. Favorable dynamic and thermodynamic conditions over the southeastern WNP promote southeastward shifted genesis and significantly longer northwestward development distances. After reaching major intensity, CE TCs travel farther westward and northward than major non-CE TCs in Warm phase, favoring continued intensification and convective axisymmetrization. These findings clarify how a climate-scale mode modulates vortex-scale structure and help anticipate periods of elevated CE-related risk.

Net Contributions of Multiple Oceanic Warm and Cold Thermal Events to Tropical Cyclone Intensity Revealed by Idealized Coupled Simulations

Authors: Wenting Lou, Tongya Liu, Jiacheng Hong, Yipeng Guo, Qingyang Xiong, Han Zhang, Changlin Chen, Jinlin Ji, and Weijie Sun.

Published in *Geoscience Letters*, 2026.

DOI: 10.1186/s40562-026-00491-0

Abstract. The ocean serves as the energy source for tropical cyclones (TCs), and a TC typically encounters multiple oceanic warm and cold events during its lifecycle. While most existing studies emphasize individual thermal events, the net contribution of multiple warm and cold events to TC intensity remains unclear. We use a set of idealized coupled atmosphere–ocean simulations to examine how multiple ocean warm and cold thermal events affect TC intensity evolution throughout its lifecycle. Based on observational evidence, our experimental design includes one control experiment and 30 sensitivity experiments across three core scenarios (warm-anomaly dominated, cold-anomaly dominated, and warm-cold balanced anomalies). Relative to the control experiment, warm-dominated experiments produce a stronger intensity evolution, cold-dominated experiments produce a weaker decay, and the balanced experiments also lead to a slight increase. This response is fundamentally asymmetric, as the intensifying influence of warm events outweighs the weakening influence exerted by cold events with comparable magnitude. Moreover, this asymmetry increases with the magnitude of the oceanic thermal anomalies, and when the warm-anomaly amplitude is doubled, the simulated TC satisfies the rapid intensification criterion. These findings quantify the asymmetric forcing of ocean thermal structures on TCs, providing a crucial mechanistic basis for improving TC intensity forecasts.

The Intrinsic Timing Relationship Between Tropical Cyclone Maximum Intensification Rate and Lifetime Maximum Intensity

Authors: Qingyang Xiong and Qiaoyan Wu.

Published in *Geophysical Research Letters*, 2025.

DOI: 10.1029/2024GL113158

Abstract. The intensification rate (IR) equation, derived from a time-dependent simplified energetically based dynamical system (EBDS), links tropical cyclone (TC) intensity, IR, and their evolutions, suggesting a connection between the timing of TC lifetime maximum intensity (LMI) and the maximum intensification rate (MIR). This study demonstrates that the IR equation predicts MIR typically preceding LMI within 36 hr, as confirmed by numerical simulation and analysis of best-track data. It was observed that the majority of global TCs reached LMIs within 36 hr after MIR from 1982 to 2023. This consistency between observational data, numerical simulations, and analytical results highlights the EBDS-based IR equation's ability to capture the timing relationship between MIR and LMI. This intrinsic link highlights the importance of understanding the timing of MIR in the evolution of TCs.